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B.Tech. Degree IV Semester Supplementary Examination April 2022

ME 15-1404 APPLIED THERMODYNAMICS (2015 Scheme)

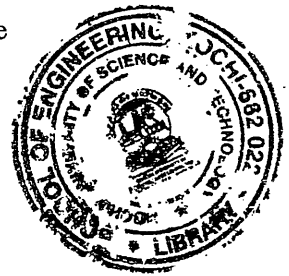
Time: 3 Hours

Maximum Marks: 60

PART A (Answer *ALL* questions)

(10 × 2 = 20)

- I. (a) Show that the first law of thermodynamics leads to the fact that energy is a property of a system.
- (b) State the two statements of second law of thermodynamics. Also show the equivalence of two statements.
- (c) Describe the concept of availability and irreversibility.
- (d) Explain P-V-T surface with the help of a neat diagram.
- (e) Describe a Reheat cycle with a neat sketch and show PV and TS diagram.
- (f) Why are supersonic nozzles convergent – divergent?
- (g) Elaborate on pressure and velocity compounded steam turbine.
- (h) Explain Dalton's law of additive pressures and Amagat's law of additive volumes.
- (i) Differentiate between characteristic gas constant and universal gas constant.
- (j) What is apparent molecular mass? Does the sum of molecular mass of all the components in a mixture give the apparent molar mass?



PART B

(4 × 10 = 40)

- II. (a) Derive steady flow energy equation. Apply SFEE to compressor, nozzle and throttling device. (4)
- (b) A reversible heat engine delivers 65 kW power and rejects energy at the rate of 40 kJ/s to a reservoir at 27°C. Determine the efficiency of the engine and the temperature at which energy is absorbed by the engine. (6)

OR

- III. (a) Derive Maxwell equations from fundamentals. (5)
- (b) Define and explain Carnot theorem. Show that the efficiency of a reversible engine is independent of the nature or amount of working substance. (5)

- IV. (a) Explain the working of a Regenerative steam power cycle. (4)
- (b) In a thermal power plant working on Rankine cycle, superheated steam at 20 bar and 400°C is produced in the boiler and the condenser is operated at 0.2 bar. Calculate the quality of the steam at the turbine outlet and the thermal efficiency of the cycle. (6)

OR

- V. (a) Explain Mercury-Water binary vapour cycle. (4)
- (b) Write a note on factors influencing Boiler Efficiency. (6)

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- VI. (a) Explain the effect of friction on nozzle. (4)
 (b) Dry saturated steam is expanded in a steam nozzle from 10 bar to 0.1 bar. Calculate: (6)
 (i) The dryness fraction of steam at exit
 (ii) Enthalpy drop
 (iii) Velocity of steam at exit if the initial velocity is 135 m/s.

OR

- VII. (a) Write a note on types and classification of steam turbines. (4)
 (b) Steam issues from the nozzles of a De-Laval turbine with a velocity of 1200 m/s. The nozzle angle is 20° , the mean blade velocity is 400 m/s and the inlet and outlet angles of blade are equal. The rate of mass flow through the turbine is 900 kg/hr. Calculate: (6)
 (i) the blade angles
 (ii) the relative velocity of steam entering the blade
 (iii) the power developed.
 Assume blade velocity co-efficient as 0.8.

- VIII. (a) Explain volumetric and gravimetric analysis of gas mixtures. (4)
 (b) A gaseous fuel with a volumetric analysis of 40% ethane, 40% butane and 20% pentane is flowing through a pipeline at the rate of $100 \text{ m}^3/\text{s}$ at 1 bar and 300 K. Calculate the partial pressure of each component and the mass flow rate. (6)

OR

- IX. (a) Derive an expression for the apparent molecular mass of a mixture of gas. (4)
 (b) A cylinder of volume 0.25 m^3 initially contains a mixture of 20% propane (C_3H_8) and 80% butane (C_4H_{10}) on molar basis, at 1 bar and 300 K. The cylinder is connected to a line carrying nitrogen at 300 K and the cylinder is charged isothermally to a pressure of 1.5 bar. Determine the mass of nitrogen that entered the cylinder. (6)
